

BTSecBench: A Comprehensive Benchmark Framework for Backdoor Attack Detection and Defense in Traffic Sign Recognition Using Deep Learning

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Abstract- The rapid deployment of deep learning-based Traffic Sign Recognition (TSR) systems has significantly improved the capabilities of autonomous driving and Advanced Driver Assistance Systems (ADAS). Despite their high predictive performance, modern convolutional neural networks remain susceptible to backdoor attacks, in which adversaries implant hidden triggers during training that cause targeted misclassification while maintaining high accuracy on clean data. Such attacks present a critical security risk because compromised models often appear trustworthy during conventional evaluation and only exhibit malicious behaviour when specific trigger patterns are present. This project presents BTSecBench, a comprehensive benchmark framework designed to evaluate the robustness of traffic sign recognition models against multiple backdoor attack strategies while integrating detection, defense, and automated benchmarking within a unified experimental pipeline. The framework employs the German Traffic Sign Recognition Benchmark (GTSRB) dataset and evaluates several deep learning architectures before selecting Efficient Net-B0 as the backbone model based on its superior balance between classification performance and computational efficiency. Four representative backdoor attacks BadNets, Blend, SIG, and WaNet-are implemented to assess the vulnerability of the selected model under diverse trigger mechanisms. To enhance system security, the STRIP entropybased detection algorithm is incorporated to identify poisoned inputs during inference, while a Fine-Tuning defense strategy is applied to recover model integrity without sacrificing clean-data performance. The framework further integrates automated evaluation modules that measure Attack Success Rate (ASR), classification accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), and benchmark summaries. Experimental results demonstrate that the proposed framework successfully integrates attack generation, detection, defense, and evaluation into a single reproducible pipeline suitable for academic research and cybersecurity education.

Index Terms - Traffic Sign Recognition, Deep Learning, Backdoor Attacks, BadNets, Blend Attack, SIG Attack, WaNet, Efficient Net-B0, STRIP, Fine-Tuning, Cybersecurity, Adversarial

TABLE OF ACRONYMS

Acronym	Meaning
TSR	Traffic Sign Recognition
GTSRB	German Traffic Sign Recognition Benchmark
CNN	Convolutional Neural Network
ASR	Attack Success Rate
MCC	Matthews Correlation Coefficient
STRIP	STRong Intentional Perturbation
ADAS	Advanced Driver Assistance Systems
AI	Artificial Intelligence
DL	Deep Learning

I. INTRODUCTION

A. Background

Traffic Sign Recognition (TSR) has become a fundamental component of intelligent transportation systems, autonomous vehicles, and Advanced Driver Assistance Systems (ADAS). Modern TSR systems employ deep convolutional neural networks capable of recognizing traffic signs with remarkable accuracy under varying environmental conditions. These systems assist vehicles in interpreting road regulations, improving driving safety, and enabling autonomous decision-making. Recent advances in deep learning architectures, particularly convolutional neural networks (CNNs), have significantly enhanced recognition performance, allowing models to achieve nearhuman accuracy on benchmark datasets such as the German Traffic Sign Recognition Benchmark (GTSRB). Despite these advances, the widespread adoption of deep learning has introduced new security challenges. Neural networks can be manipulated through adversarial attacks that compromise model behaviour without noticeably affecting standard performance metrics. Among these threats, backdoor attacks have emerged as one of the most dangerous because they remain dormant during normal operation while activating only when a predefined trigger is present. Unlike traditional adversarial examples, backdoor attacks permanently alter the model during training, enabling attackers to control predictions while maintaining apparently normal accuracy on clean data.

B. Motivation

The security of machine learning models is becoming increasingly important as artificial intelligence systems are deployed in safety-critical applications. Traffic sign recognition models influence real-world driving decisions, making them attractive targets for malicious manipulation. A successful backdoor attack could force an autonomous vehicle to misinterpret critical traffic signs, potentially resulting in unsafe driving behaviour and severe consequences. Most existing academic studies evaluate individual attack or defense mechanisms independently. However, there is a lack of unified

benchmarking frameworks that systematically compare multiple attacks, evaluate detection algorithms, assess defense strategies, and generate reproducible experimental reports within a single pipeline. This limitation motivated the development of BTSecBench, which provides an integrated framework for benchmarking the security of traffic sign recognition systems under various backdoor scenarios.

C. Problem Statement

Deep learning models trained for traffic sign recognition are vulnerable to hidden backdoor attacks that can manipulate predictions while preserving high classification accuracy on clean images. Conventional evaluation methods primarily focus on overall accuracy and therefore fail to reveal hidden malicious behaviours embedded within compromised models. Consequently, there is a need for a systematic framework capable of evaluating backdoor attacks, detecting poisoned inputs, applying defense mechanisms, and measuring security performance using comprehensive evaluation metrics.

D. Objectives

The primary objectives of this project are: 1) Develop a high-performance traffic sign recognition model using the GTSRB dataset. 2) Compare multiple deep learning architectures and select the most suitable backbone model. 3) Implement representative backdoor attacks, including BadNets, Blend, SIG, and WaNet. 4) Evaluate attack effectiveness using Attack Success Rate (ASR) and standard classification metrics. 5) Integrate the STRIP entropy-based algorithm for runtime backdoor detection. 6) Implement a Fine-Tuning defense strategy to mitigate backdoor behaviour while maintaining clean-data accuracy. 7) Design an automated benchmarking framework capable of generating reports, visualizations, and comparative analyses.

E. Contributions

The major contributions of BTSecBench are summarized as follows: • Development of a modular benchmark framework for evaluating backdoor security in traffic sign recognition. • Comparative evaluation of multiple deep learning architectures leading to the selection of Efficient Net-B0 as the backbone network. • Implementation of four representative backdoor attack techniques within a unified experimental pipeline. • Integration of STRIP for entropy-based backdoor detection and Fine-Tuning for model recovery. • Automated generation of benchmark reports, evaluation metrics, and visualization figures to support reproducible experimentation.

F. Report Organization

The remainder of this report is organized as follows. Section II reviews related work on traffic sign recognition, deep learning architectures, and backdoor attacks. Section III presents the problem statement and project objectives. Section IV describes the dataset and exploratory data analysis. Section V details the proposed methodology and system architecture. Section VI compares the evaluated deep learning models and justifies the selection of Efficient Net-B0. Section VII explains the implementation of backdoor attacks. Section VIII discusses the STRIP detection mechanism, while Section IX presents the Fine-Tuning defense strategy. Section X summarizes the experimental results and benchmark analysis. Section XI provides an error analysis and discusses the limitations of the proposed framework. Finally, Section XII concludes the report and outlines future research directions.

II. LITERATURE REVIEW

A. Introduction

Traffic Sign Recognition (TSR) has become one of the most important perception tasks in intelligent transportation systems and autonomous driving. Recent advances in deep learning have enabled

CNN-based models to achieve remarkable classification accuracy on benchmark datasets such as the German Traffic Sign Recognition Benchmark (GTSRB). Despite these improvements, deep neural networks remain vulnerable to various security threats, including adversarial examples, data poisoning, and backdoor attacks. These vulnerabilities have raised significant concerns regarding the deployment of deep learning models in safety-critical applications where incorrect predictions may result in severe consequences. Unlike conventional image classification research, security-oriented machine learning focuses not only on improving predictive performance but also on ensuring robustness against malicious attacks. Consequently, recent studies have investigated both attack strategies and defense mechanisms to improve the trustworthiness of deep learning systems.

B. Traffic Sign Recognition Using Deep Learning

Traditional traffic sign recognition systems relied on handcrafted image descriptors such as Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), Haar features, and Support Vector Machines (SVMs). Although these approaches achieved reasonable performance under controlled conditions, they often struggled with variations in illumination, occlusion, motion blur, weather conditions, and camera viewpoints. The introduction of Convolutional Neural Networks (CNNs) significantly improved feature extraction by automatically learning hierarchical image representations directly from raw pixel values. CNN-based methods eliminate the need for handcrafted features and demonstrate superior robustness across diverse environmental conditions. Several benchmark studies have shown that modern CNN architectures consistently outperform traditional machine learning approaches on the 98%.

C. Evolution of Deep Learning Architectures

Deep learning architectures have evolved rapidly over the past decade. Early CNN models focused primarily on increasing network depth, whereas modern architectures emphasize computational efficiency and parameter optimization.

TABLE I: Dataset Characteristics

Characteristic	Value
Dataset	GTSRB
Number of Classes	43
Image Format	RGB
Image Size	Variable, resized to 224x224
Learning Task	Image Classification

1) Baseline CNN: A conventional CNN typically consists of convolutional layers followed by pooling operations and fully connected classifiers. Such architectures are relatively simple to implement and provide a useful benchmark for comparison. However, increasing network depth often leads to optimization difficulties, limiting their ability to capture complex visual patterns. Advantages: simple implementation, low computational complexity, and easy interpretability. Limitations: lower representational capability, reduced generalization performance, and difficulty learning complex traffic sign variations. 2) Res Net18: Residual Networks (Res Net) introduced skip connections that allow gradients to propagate effectively through deep architectures. This innovation addressed the degradation problem encountered in very deep neural networks and enabled substantially deeper models to be trained successfully. Res Net18 has become one of the most widely adopted baseline architectures due to its stable convergence and strong feature extraction capability. Advantages: excellent feature learning, stable optimization, and strong generalization. Limitations: higher computational cost and larger memory footprint. 3) Mobile Net V3: Mobile Net V3 was specifically designed for resource-constrained devices such as smartphones and embedded systems. It combines

depthwise separable convolutions, squeeze-and-excitation modules, and neural architecture search to achieve excellent computational efficiency. Advantages: lightweight architecture, fast inference, and low memory consumption. Limitation: slightly reduced feature representation compared with larger architectures. 4) Efficient Net-B0: Efficient Net introduced compound scaling, simultaneously scaling network depth, width, and image resolution using a mathematically derived scaling strategy. Among the Efficient Net family, Efficient Net-B0 provides an excellent compromise between computational efficiency and classification accuracy, making it particularly suitable for security-oriented benchmark frameworks. Advantages: high classification accuracy, parameter efficiency, excellent computational performance, and strong generalization capability. Because of these characteristics, Efficient Net-B0 was selected as the backbone architecture for all subsequent experiments within BTSecBench.

D. Backdoor Attacks in Deep Learning

Backdoor attacks represent one of the most serious security threats affecting deep neural networks. Unlike adversarial examples that manipulate inputs only during inference, backdoor attacks modify the training process itself by injecting poisoned samples containing predefined trigger patterns. The objective of a backdoor attack is twofold: to maintain high classification accuracy on clean images, and to force targeted misclassification whenever a trigger appears. This makes backdoor attacks particularly dangerous because compromised models often pass conventional evaluation procedures. 1) BadNets: BadNets is considered the first practical backdoor attack against deep neural networks. It introduces a visible trigger patch into selected training images while changing their labels to a predefined target class. Once trained, the model behaves normally on clean images but predicts the attacker-selected class whenever the trigger is present. Advantages: high attack success rate and simple implementation. Limitation: the trigger may be visually detectable. 2) Blend Attack: Blend Attack improves upon BadNets by blending trigger patterns with the original image using transparency coefficients. Because the trigger becomes partially invisible, Blend attacks are considerably more difficult to detect using visual inspection. Advantages: less visible trigger and high stealth. Limitation: slightly lower attack effectiveness under certain settings. 3) SIG Attack: The Sinusoidal Signal (SIG) attack introduces a global sinusoidal perturbation distributed across the entire image rather than placing a localized trigger patch. The distributed nature of the trigger makes manual inspection substantially more difficult while maintaining effective attack performance. Advantages: global trigger and difficult visual detection. Limitation: more complex trigger generation. 4) WaNet: WaNet represents one of the most sophisticated backdoor attacks currently available. Instead of modifying image intensities directly, WaNet applies subtle geometric warping transformations that preserve natural image appearance. This spatial transformation significantly improves attack stealth while maintaining high attack effectiveness. Advantages: nearly invisible trigger, strong attack capability, and high stealth. Limitations: complex implementation and increased computational cost.

E. Detection of Backdoor Attacks

Numerous defense mechanisms have been proposed to identify poisoned samples during inference. Existing approaches include activation clustering, neural cleanse, spectral signatures, fine-pruning, and entropy-based methods. Among these techniques, STRIP (STRong Intentional Perturbation) has emerged as one of the most practical runtime detection mechanisms. Instead of inspecting model parameters, STRIP evaluates prediction entropy by repeatedly perturbing incoming images with random overlays. Clean inputs typically exhibit high prediction uncertainty, whereas poisoned inputs consistently activate the embedded trigger and therefore produce unusually low entropy. This difference allows STRIP to distinguish

between benign and poisoned inputs without requiring modifications to the underlying neural network. Advantages: model-independent, runtime detection, no retraining required. Limitations: threshold selection influences performance, and detection effectiveness depends on trigger characteristics.

F. Defense Against Backdoor Attacks

Backdoor defense techniques can generally be categorized into three groups: data-based defenses, model-based defenses, and runtime detection mechanisms. Data-based defenses attempt to identify poisoned samples before training. Modelbased defenses modify network parameters to suppress malicious behavior. Runtime approaches inspect incoming inputs during inference. Among model-based defenses, Fine-Tuning has become one of the simplest and most effective approaches for recovering compromised models. By retraining only the final layers using clean data, Fine-Tuning gradually weakens the association between trigger patterns and attacker-selected target classes while preserving the original classification capability. This project adopts Fine-Tuning as the primary defense mechanism due to its simplicity, computational efficiency, and compatibility with the selected Efficient Net-B0 architecture.

G. Research Gap

Although numerous studies investigate individual backdoor attacks or defense techniques, most existing work evaluates these components independently. The following limitations are commonly observed: • Evaluation restricted to a single attack. • Lack of standardized benchmarking. • Limited comparison of multiple deep learning architectures. • Minimal integration of detection and defense. • Insufficient automated reporting and reproducibility. To address these shortcomings, BTSecBench proposes a unified benchmark framework that integrates model selection, multiple backdoor attacks, STRIP detection, Fine-Tuning defense, comprehensive evaluation metrics, automated benchmark generation, and visualization into a single reproducible experimental pipeline.

III. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

A. Problem Statement

Recent advancements in Deep Learning (DL) have significantly improved the performance of Traffic Sign Recognition (TSR) systems used in autonomous vehicles and Advanced Driver Assistance Systems (ADAS). Modern convolutional neural networks are capable of recognizing traffic signs with near-human accuracy under diverse environmental conditions, making them a fundamental component of intelligent transportation systems. However, despite these remarkable improvements, deep learning models remain vulnerable to a variety of security threats that can compromise their reliability and safety. Among these threats, backdoor attacks represent one of the most dangerous forms of data poisoning. Unlike traditional adversarial attacks that modify only the inference input, backdoor attacks manipulate the training dataset by injecting poisoned samples containing hidden trigger patterns. During training, the model unintentionally learns an association between the trigger and a target label while maintaining high classification accuracy on clean images. Consequently, the compromised model behaves normally during conventional evaluation but produces attacker-controlled predictions whenever the trigger is present.

TABLE II: Comparison of Evaluated Deep Learning Architectures

Model	Params (M)	Efficiency	Selected
Baseline CNN	Varies	Moderate	No
ResNet18	11.7	Good	No
MobileNetV3	5.4	Excellent	No
EfficientNet-B0	5.3	Excellent	Yes

TABLE III: EfficientNet-B0 Validation Performance

Metric	Value
Validation Accuracy	99.99%
Precision	99.99%
Recall	99.99%
F1-Score	99.99%
Balanced Accuracy	99.99%
MCC	0.9999

This vulnerability is particularly critical in traffic sign recognition systems because incorrect classification of road signs may directly affect vehicle decision-making. For example, an attacker could manipulate a Stop sign to be classified as a Speed Limit sign, potentially leading to unsafe driving behavior. Since conventional evaluation metrics primarily measure overall classification accuracy, they often fail to reveal hidden malicious behavior embedded within trained models. Existing research has proposed numerous backdoor attacks, including BadNets, Blend, SIG, and WaNet, as well as several defense techniques such as Neural Cleanse, STRIP, Fine Pruning, and Fine-Tuning. However, most studies evaluate these techniques independently, making it difficult to compare attacks, detection methods, and defense mechanisms within a unified experimental framework. Furthermore, many implementations lack automated benchmarking, standardized evaluation protocols, and reproducible reporting, limiting their usefulness for educational and comparative research. Therefore, there is a clear need for a comprehensive benchmark framework capable of systematically evaluating multiple backdoor attacks, runtime detection methods, defense strategies, and performance metrics while maintaining reproducibility and ease of experimentation.

B. Research Motivation

The increasing deployment of deep learning models in safety-critical applications has shifted the focus of machine learning research from achieving high predictive accuracy toward developing secure, robust, and trustworthy artificial intelligence systems. Autonomous driving is one of the most prominent examples where incorrect model predictions may have significant real-world consequences. Traffic Sign Recognition serves as a critical perception module within autonomous driving pipelines. Any successful compromise of the recognition model can directly influence vehicle control decisions, making TSR systems attractive targets for adversarial manipulation. While numerous studies investigate either attack strategies or defense mechanisms individually, relatively few provide a complete framework for analyzing the interaction between multiple attacks and defenses under standardized evaluation conditions. The motivation behind BTSecBench is therefore to bridge this gap by developing an integrated benchmarking framework that enables researchers and students to evaluate the robustness of traffic sign recognition models using multiple attack scenarios, runtime detection algorithms, and recovery techniques within a single modular pipeline. In addition to improving reproducibility, the framework aims to simplify comparative analysis by automating report generation, visualization, and benchmark creation.

C. Research Objectives

The primary objective of this research is to design and implement a comprehensive benchmark framework for evaluating the security of deep learning-based traffic sign recognition systems against multiple backdoor attacks. The specific objectives of the project are as follows: 1) Develop a reliable Traffic Sign Recognition model using the German Traffic Sign Recognition Benchmark (GTSRB) dataset. 2) Compare multiple deep learning architectures, including a Baseline CNN, Res Net18, Mobile Net V3, and Efficient Net-B0, to identify the most suitable backbone network for security evaluation. 3)

Implement four representative backdoor attack techniques-BadNets, Blend, SIG, and WaNet-to evaluate different trigger generation strategies and attack characteristics. 4) Measure attack effectiveness using Attack Success Rate (ASR) alongside conventional classification metrics such as accuracy, precision, recall, F1-score, balanced accuracy, and Matthews Correlation Coefficient (MCC). 5) Integrate the STRIP (STRong Intentional Perturbation) algorithm to perform runtime detection of poisoned inputs using entropy-based analysis. 6) Implement a Fine-Tuning defense strategy to recover compromised models while preserving clean-data classification performance. 7) Develop an automated benchmarking framework capable of generating benchmark reports, visualization figures, and comparative performance summaries. 8) Evaluate the overall robustness of the proposed framework by combining attack implementation, detection, defense, and benchmarking within a reproducible experimental workflow.

D. Scope of the Project

The scope of BTSecBench focuses on supervised image classification using the German Traffic Sign Recognition Benchmark (GTSRB) dataset. The project investigates the security of deep learning models against backdoor attacks rather than general adversarial robustness. The benchmark framework includes four major components: selection of the optimal deep learning backbone through comparative evaluation, implementation of multiple backdoor attack algorithms, runtime detection using STRIP, and model recovery using Fine-Tuning. The framework additionally provides automated performance evaluation and benchmark generation to facilitate reproducible experimentation. Although the project investigates several representative attack techniques, it does not attempt to evaluate every existing backdoor attack or defense algorithm. Similarly, the proposed defense focuses on Fine-Tuning and does not include more computationally intensive approaches such as Neural Cleanse or Spectral Signature removal. These techniques are identified as potential extensions for future research.

E. Research Questions

The project seeks to answer the following research questions: • RQ1: Which deep learning architecture provides the best balance between classification performance and computational efficiency for traffic sign recognition? • RQ2: How vulnerable is the selected traffic sign recognition model to different backdoor attack techniques? • RQ3: Can STRIP effectively distinguish poisoned inputs from clean traffic sign images during runtime? • RQ4: To what extent can Fine-Tuning reduce the influence of embedded backdoor triggers while maintaining high classification accuracy? • RQ5: Can multiple attack, detection, and defense mechanisms be integrated into a unified benchmark framework that supports reproducible security evaluation?

F. Expected Contributions

- A modular and extensible benchmark framework for backdoor security research.
- Comparative evaluation of multiple deep learning architectures for traffic sign recognition.
- Unified implementation of four representative backdoor attacks.
- Integration of STRIP as a runtime detection mechanism.
- Implementation of Fine-Tuning as a practical defense strategy.
- Automated benchmark generation supporting CSV, JSON, Excel, Markdown, and publication-quality visualizations.
- A reproducible workflow suitable for academic research, cybersecurity education, and comparative evaluation of backdoor robustness.

IV. DATASET AND EXPLORATORY DATA ANALYSIS

A. Introduction

The quality and diversity of the dataset play a critical role in determining the performance and robustness of deep learning models. Since BTSecBench focuses on evaluating the security of Traffic Sign

Recognition (TSR) systems under multiple backdoor attack scenarios, selecting an appropriate benchmark dataset is essential for ensuring realistic experimentation and meaningful evaluation. This study employs the German Traffic Sign Recognition Benchmark (GTSRB), one of the most widely used public datasets for traffic sign classification. Unlike synthetic datasets, GTSRB contains substantial variability in illumination, viewing angle, scale, motion blur, weather conditions, and partial occlusions, making it representative of practical autonomous driving scenarios.

B. German Traffic Sign Recognition Benchmark (GTSRB)

The German Traffic Sign Recognition Benchmark (GTSRB) was introduced as a standardized dataset for evaluating machine learning algorithms designed for traffic sign recognition. The dataset contains 43 distinct traffic sign classes, covering a wide range of regulatory, warning, mandatory, and prohibitory signs encountered on German roads. Images were captured using vehicle-mounted cameras under real driving conditions, resulting in substantial variations in viewpoint, background clutter, lighting conditions, and object scale. Each image is

TABLE IV: Comparison of Implemented Backdoor Attacks

Attack	Trigger Type	Visibility
BadNets	Local patch	High
Blend	Blended overlay	Medium
SIG	Global sinusoidal	Low
WaNet	Warping field	Low

associated with a single ground-truth label, and the dataset is organized into predefined training and testing subsets. Property Value Classification Type Multi-class

C. Dataset Selection Rationale

The GTSRB dataset was selected for several reasons. First, it represents one of the most extensively studied datasets for traffic sign recognition, enabling direct comparison with previous research. Second, the large number of traffic sign categories provides sufficient complexity for evaluating modern deep learning architectures. Third, the real-world acquisition conditions introduce significant visual variability that better reflects practical deployment environments. Most importantly, machine learning research, making it well suited for evaluating backdoor attacks and defense mechanisms.

D. Exploratory Data Analysis (EDA)

Before developing the deep learning models, an extensive exploratory analysis of the dataset was conducted to understand its characteristics, verify data quality, identify potential class imbalance, and determine appropriate preprocessing strategies. The exploratory analysis consisted of the following steps: • Inspection of the directory structure. • Verification of class labels. • Analysis of dataset size. • Examination of representative image samples. • Identification of image resolution variations. • Assessment of class distribution. • Validation of training and testing subsets.

E. Dataset Distribution

The dataset contains images belonging to 43 traffic sign categories, each representing a unique road sign type. Although the overall dataset is sufficiently large for deep learning applications, the number of samples available for individual classes varies considerably. Certain frequently encountered traffic signs contain substantially more training examples than rare categories, resulting in moderate class imbalance, a common characteristic of real-world traffic datasets. Rather than artificially balancing the dataset through extensive oversampling or undersampling, this study retains the natural class distribution to preserve realistic training conditions, and instead relies on data augmentation to improve generalization.

Fig. 1: Class distribution of the GTSRB dataset.

F. Sample Image Analysis

Representative samples from each traffic sign category were manually inspected to understand the visual characteristics of the dataset. Several observations were made: traffic signs appear at different scales due to varying camera distances; images exhibit substantial illumination changes caused by weather and time-of-day variations; motion blur is present in several images captured at higher vehicle speeds; partial occlusions occasionally occur because of surrounding objects such as trees, vehicles, or pedestrians; background clutter varies significantly across scenes; and certain traffic signs possess highly similar visual appearances, increasing classification difficulty. These observations justify the use of deep convolutional neural networks capable of learning robust hierarchical feature representations.

Fig. 2: Representative traffic sign samples from different categories.

G. Data Preprocessing

To ensure consistent input representation across all experiments, several preprocessing operations were applied before training the neural networks. Image Resizing. Original GTSRB images possess varying spatial resolutions. Therefore, all images were resized to 224x224 pixels, matching the input requirements of Efficient Net-B0 and the other evaluated CNN architectures. Normalization. Pixel intensities were normalized using the mean and standard deviation computed from the training dataset, improving numerical stability during optimization and accelerating network convergence. Data Augmentation. To improve model robustness and reduce overfitting, online data augmentation was employed during training. The augmentation pipeline included random horizontal transformations (where appropriate), random rotations, random translations, random scaling, color jitter, and random affine transformations. These augmentations expose the network to a wider variety of visual appearances without modifying the underlying semantic labels.

H. Dataset Splitting Strategy

The dataset was divided into three subsets: a training set used for model learning and parameter optimization; a validation set used for hyperparameter tuning and model selection; and a test set used exclusively for final performance evaluation. Maintaining independent validation and testing subsets prevents information leakage and provides an unbiased estimate of model generalization. The same data split was consistently used across all evaluated architectures and backdoor attack experiments, ensuring fair comparison between different methods.

I. Data Quality Assessment

Prior to training, the dataset was examined for potential quality issues. The following checks were performed: verification of image readability, confirmation of label consistency, detection of corrupted files, inspection of duplicate images, validation of folder structure, and confirmation of successful preprocessing. No significant inconsistencies affecting the experimental pipeline were observed.

J. Importance of Dataset Preparation

The reliability of deep learning models depends heavily on the quality of the training data. Proper preprocessing not only improves classification accuracy but also ensures that subsequent backdoor attack experiments accurately evaluate model robustness rather than being influenced by dataset artifacts. Since BTSecBench investigates both security attacks and defense mechanisms, maintaining a consistent preprocessing pipeline across all experiments is essential for ensuring reproducibility and fair comparison.

V. METHODOLOGY

A. Overview of the Proposed Framework

This section presents the methodology adopted for developing BTSecBench, a comprehensive benchmark framework for evaluating the robustness of deep learning-based Traffic Sign Recognition (TSR) systems against backdoor attacks. The proposed framework follows an end-to-end security evaluation pipeline consisting of five major stages: (1) deep learning model development and selection; (2) backdoor attack generation; (3) backdoor detection; (4) defense and recovery; and (5) automated benchmarking and reporting. Unlike conventional machine learning workflows that focus only on classification accuracy, BTSecBench evaluates both performance and security robustness.

B. Overall System Architecture

The proposed BTSecBench architecture processes the GTSRB dataset through a data preprocessing pipeline, followed by deep learning model selection among Baseline CNN, Res Net18, Mobile Net V3, and Efficient Net-B0. The selected Efficient Net-B0 backbone is then subjected to backdoor attack generation using BadNets, Blend, SIG, and WaNet, followed by backdoor evaluation via STRIP detection and Fine-Tuning defense. The pipeline concludes with automated benchmark and report generation in CSV, JSON, Excel, and figure for-

TABLE V: STRIP Detection Performance

Metric	Value
Detection Accuracy	56.61%
Precision	53.88%
Recall	91.70%
F1-Score	67.88%
ROC-AUC	0.528

TABLE VI: Fine-Tuning Configuration

Setting	Value
Learning Strategy	Fine-Tuning
Backbone	EfficientNet-B0
Input Size	224x224
Evaluation Metrics	Accuracy, Precision, Recall, F1, MCC, ASR

TABLE VII: Fine-Tuning Performance

Metric	Value
Validation Accuracy	99.94%
Precision	99.87%
Recall	99.93%
F1-Score	99.90%
MCC	0.9993
Validation Loss	0.004

mats.

Fig. 3: Overall system architecture of BTSecBench.

C. Model Development Pipeline

1) Baseline Model: The first stage of the methodology involved developing a baseline convolutional neural network, which provides a reference point for evaluating improvements achieved through more advanced architectures. The purpose of the baseline model is to establish initial classification performance, understand dataset difficulty, provide a comparison against advanced architectures, and demonstrate the benefit of modern CNN designs. The baseline architecture consists of convolutional feature extraction layers, activation functions, pooling layers, and fully connected classification layers. 2) Advanced Model Evaluation: To identify the optimal backbone network, multiple deep learning architectures were evaluated: Baseline CNN, Res Net18, Mobile Net V3, and Efficient Net-B0. The purpose of this comparison was not only to maximize

accuracy but also to identify a model suitable for security evaluation, requiring high classification accuracy, efficient parameter usage, strong feature representation, and reasonable computational requirements.

D. Res Net18 Architecture

Res Net18 introduces residual connections that address the degradation problem encountered in deeper neural networks. Traditional deep networks suffer from vanishing gradients, difficult optimization, and reduced training stability. Residual learning solves this by introducing shortcut connections: $y = F(x) + x$ (1) where x represents the original input, $F(x)$ represents the learned residual mapping, and y represents the final output. The residual structure allows information to propagate efficiently through the network, improving training stability. Advantages: strong feature extraction, stable convergence, and a well-established benchmark architecture. Limitation: a higher parameter count compared with lightweight architectures.

E. Mobile Net V3 Architecture

Mobile Net V3 was designed specifically for efficient deployment on resource-constrained devices. The architecture uses depthwise separable convolutions, squeeze-and-excitation blocks, and hardware-aware optimization. A standard convolution operation can be decomposed into a depthwise convolution followed by a pointwise convolution, significantly reducing computational cost while maintaining competitive accuracy. Advantages: lightweight design, reduced memory usage, and fast inference. Limitation: lower feature capacity compared with larger networks.

F. Efficient Net-B0 Architecture

Efficient Net-B0 was selected as the final backbone architecture for BTSecBench. Unlike conventional CNN scaling approaches that independently increase depth, width, or resolution, Efficient Net introduces compound scaling, represented as: Depth = $\alpha\phi$, Width = $\beta\phi$, Resolution = $\gamma\phi$ (2) where ϕ controls the overall scaling factor, and α , β , and γ control depth, width, and resolution, respectively. This approach enables Efficient Net to achieve higher accuracy with fewer parameters. The advantages of Efficient Net-B0 include a superior accuracy-efficiency trade-off, a compact architecture, strong generalization, and reduced computational requirements. Based on experimental evaluation, Efficient Net B0 achieved the best classification performance and was therefore selected as the backbone model for all subsequent backdoor security experiments.

G. Training Methodology

All models were trained using a consistent experimental setup to ensure fair comparison. Loss Function. Multi-class classification was optimized using Cross-Entropy Loss: $C = -\sum_{i=1}^C y_i \log(\hat{y}_i)$ (3) where C represents the number of classes, y_i represents the true label, and $\hat{y}_i$ represents the predicted probability. Optimizer. The Adam optimizer was selected due to its adaptive learning rate adjustment, faster convergence, and robust optimization performance. Learning Strategy. The training pipeline included batchbased optimization, validation monitoring, best-model checkpoint saving, and performance tracking. The best-performing checkpoint was preserved for later attack, detection, and defense evaluation.

H. Backdoor Security Evaluation Pipeline

After selecting Efficient Net-B0, the model was evaluated under malicious scenarios. The security evaluation pipeline consisted of three steps: generating poisoned datasets, in which attack algorithms modify selected training samples by adding triggers; training compromised models, in which the model learns both normal classification behaviour and trigger-totarget association; and evaluating attack effectiveness, in which images are classified into the attacker-selected target class: Successful Target Predictions Total Triggered Samples

I. Detection Methodology

BTSecBench incorporates STRIP as a runtime detection mechanism. The detection process receives an input image, applies intentional perturbations, generates multiple predictions, calculates prediction entropy, and compares entropy with a threshold. Entropy is calculated as: $(cid:88) H(x) = - p(x) \log p(x)$ (5) Backdoored inputs typically produce lower entropy because the trigger forces consistent predictions.

J. Defense Methodology

Fine-Tuning was selected as the primary defense mechanism. The process consists of loading the compromised model, removing or weakening the trigger association through additional clean-data training, and evaluating the recovered model. The defense objective is to reduce ASR, preserve clean accuracy, and maintain model usability.

K. Automated Benchmarking Framework

A major contribution of BTSecBench is automated evaluation. The framework automatically generates quantitative reports in CSV, JSON, Excel, and Markdown formats; performance metrics including accuracy, precision, recall, F1-score, MCC, ASR, and detection metrics; and visualizations including accuracy comparison, ASR comparison, STRIP metrics, and benchmark summary. This automation improves reproducibility and reduces manual errors during experimentation.

VI. DEEP LEARNING MODEL COMPARISON

A. Introduction

Selecting an appropriate backbone architecture is one of the most important decisions in developing a reliable traffic sign recognition system. The chosen model must not only achieve high classification accuracy but also provide efficient computation, good generalization capability, and sufficient robustness for subsequent backdoor attack and defense experiments. To identify the most suitable architecture for BTSecBench, four convolutional neural network models were evaluated: Baseline CNN, Res Net18, Mobile Net V3, and Efficient Net B0. The comparison considered classification performance together with computational efficiency, architectural complexity, and suitability for cybersecurity benchmarking.

B. Evaluated Deep Learning Architectures

1) Baseline CNN: The baseline model consists of a conventional convolutional neural network containing multiple convolutional layers followed by pooling operations and fully connected classification layers, used to establish a reference performance level for the GTSRB dataset. Advantages: simple implementation, low computational complexity, easy to train and interpret. Limitations: limited feature extraction capability, lower representational power, and reduced generalization under challenging conditions. 2) Res Net18: Res Net18 introduces residual learning through shortcut connections, allowing significantly deeper networks to be optimized effectively, providing stronger feature learning and improved training stability compared with the baseline CNN. Advantages: residual learning, stable opti-

TABLE VIII: Clean Classification Performance

Metric	Value
Validation Accuracy	99.99%
Precision	99.99%
Recall	99.99%
F1-Score	99.99%
Balanced Accuracy	99.99%
MCC	0.9999

TABLE IX: STRIP Detection Performance

Metric	Value
Detection Accuracy	56.61%
Precision	53.88%
Recall	91.70%
F1-Score	67.88%
ROC-AUC	0.528

TABLE X: Fine-Tuning Performance

Metric	Value
Validation Accuracy	99.94%
Precision	99.87%
Recall	99.93%
F1-Score	99.90%
MCC	0.9993
Validation Loss	0.004

mization, strong generalization. Limitations: larger parameter count and higher computational requirements. 3) Mobile Net V3: Mobile Net V3 was designed specifically for lightweight computer vision applications, combining depthwise separable convolutions with squeeze-and-excitation modules to achieve excellent computational efficiency while maintaining competitive classification accuracy. Advantages: lightweight architecture, fast inference, low memory consumption. Limitation: lower representational capacity than larger architectures for some complex tasks. 4) Efficient Net-B0: Efficient Net-B0 employs compound scaling to simultaneously optimize network depth, width, and input resolution. Unlike conventional scaling strategies, compound scaling improves accuracy without dramatically increasing computational complexity. This architecture demonstrated the strongest overall performance during model selection and was therefore selected as the backbone model for all subsequent security experiments within BTSecBench.

C. Architectural Comparison

architectural characteristics together with the measured results retained for the final selected model. tures Res Net18 11.7 Good No Mobile Net V3 5.4 Excellent No Efficient Net-B0 5.3 Excellent Yes

D. Performance of the Selected Model

After comparative evaluation, Efficient Net-B0 was selected as the backbone architecture due to its superior classification performance and computational efficiency. The final trained model achieved the following validation performance. Validation Time 27.79 s These results demonstrate that Efficient Net-B0 provides excellent predictive performance while maintaining computational efficiency, making it an appropriate backbone for evaluating security mechanisms.

E. Model Selection Criteria

The final model was selected using the following criteria: 1) Classification Performance: the model should achieve high accuracy and balanced performance across all traffic sign categories. 2) Computational Efficiency: efficient training and inference are important for reproducible benchmarking. 3) Parameter Efficiency: compact architectures reduce memory requirements while maintaining strong performance. 4) Generalization Capability: the model should perform consistently under varying image conditions and support subsequent attack and defense experiments. 5) Compatibility with Security Evaluation: the architecture should integrate seamlessly with backdoor attacks, marking. Efficient Net-B0 satisfied all of these requirements and therefore served as the backbone architecture for the remainder of the project.

F. Comparative Discussion

The comparison demonstrates the evolution from a conventional baseline CNN to increasingly sophisticated deep learning architectures. While the baseline CNN establishes a reference for performance, Res Net18 improves feature learning through residual connections, and Mobile Net V3 emphasizes computational efficiency for lightweight deployment. Efficient Net-B0 combines the advantages of these approaches by providing excellent feature extraction with relatively few parameters through compound scaling. Consequently, Efficient Net-B0 was selected as the final backbone network and used throughout the implementation of BadNets, Blend, SIG, and WaNet attacks, STRIP-based runtime detection, Fine Tuning defense, and the automated benchmarking framework.

VII. BACKDOOR ATTACK IMPLEMENTATION

A. Introduction

Backdoor attacks represent one of the most severe security threats affecting deep learning systems. Unlike conventional adversarial attacks that perturb input samples only during inference, backdoor attacks compromise the training process itself by injecting poisoned samples into the training dataset. During optimization, the neural network learns a hidden association between a predefined trigger pattern and an attacker-selected target class while simultaneously preserving high classification performance on clean images. To comprehensively evaluate the robustness of traffic sign recognition systems, BTSecBench implements four representative backdoor attack techniques with increasing levels of sophistication: BadNets, Blend, SIG, and WaNet. These attacks were selected because they represent localized, blended, global, and geometric trigger strategies, respectively, enabling a broad assessment of model vulnerability.

B. Backdoor Attack Pipeline

The general workflow adopted in BTSecBench begins with the clean GTSRB dataset, followed by trigger generation, poison sample creation, and label replacement to produce the poisoned dataset. The poisoned dataset is then used for model training, yielding a backdoored model whose Attack Success Rate is subsequently evaluated. Each implemented attack follows this common workflow while differing in the trigger generation strategy.

Fig. 4: Overall backdoor attack pipeline.

C. BadNets Attack

1) Overview: BadNets is one of the earliest and most influential backdoor attacks proposed for deep neural networks. The attack inserts a small trigger patch into selected training images and simultaneously changes their labels to an attacker-defined target class. During inference, the trained model behaves normally for clean images but predicts the target class whenever the trigger is present. 2) Trigger Design: The trigger consists of a small highcontrast patch positioned in a fixed image location (for example, the lower-right corner). It is characterized as a localized trigger with high visual contrast, a fixed spatial location, and simple implementation. 3) Poisoning Process: The poisoning procedure consists of four stages: selecting a subset of training images, inserting the trigger patch, replacing the original labels with the target class, and retraining the classifier using the poisoned dataset. The resulting model simultaneously learns normal classification and trigger activation. 4) Advantages and Limitations: Advantages: easy implementation, high attack effectiveness, and strong reproducibility. Limitations: the trigger is visually noticeable and easier to detect compared with modern attacks.

D. Blend Attack

1) Overview: The Blend Attack improves the stealthiness of conventional trigger-based attacks by blending the trigger with the original image rather than inserting a visible patch. Instead of replacing image pixels directly, the trigger is combined with the clean image using a transparency coefficient. The poisoned image can be

expressed as: $I_p = (1 - \alpha)I + \alpha T$ (6) where I_p is the poisoned image, I is the clean image, T is the trigger pattern, and α is the blending coefficient. 2) Trigger Characteristics and Trade-offs: The trigger is distributed across the image while remaining visually unobtrusive. Compared with BadNets it is less visible, offers higher stealth, and is harder for humans to identify. Advantages: high visual imperceptibility, preservation of image appearance, and effective trigger activation. Limitations: slightly more complex implementation, and performance depends on the selected blending coefficient.

E. SIG Attack

1) Overview: SIG (Sinusoidal Signal) Attack introduces a global sinusoidal perturbation across the entire image. Unlike localized trigger attacks, SIG distributes the trigger over all pixels, making it substantially more difficult to identify through manual inspection. The perturbation is represented as: $S(x, y) = A \sin(2\pi f x + \phi)$ (7) where A is the amplitude, f is the frequency, and ϕ is the phase. 2) Characteristics and Trade-offs: Unlike BadNets, SIG produces no visible patch, distributes the trigger globally, and introduces smooth intensity variation. Advantages: highly stealthy, difficult visual detection, and consistent attack activation. Limitations: requires careful parameter selection and is more computationally expensive than localized attacks.

F. WaNet Attack

1) Overview: WaNet (Warping-based Backdoor Attack) represents one of the most advanced backdoor attacks implemented within BTSecBench. Instead of modifying pixel intensities, WaNet introduces subtle geometric transformations using image warping. The resulting trigger is almost invisible to human observers while remaining highly effective. 2) Trigger Generation and Trade-offs: The attack creates a smooth displacement field that warps image coordinates. Rather than adding new visual content, existing pixels are slightly displaced, preserving the natural appearance of traffic signs while embedding the malicious trigger. Advantages: nearly invisible trigger, extremely high stealth, and strong attack capability. Limitations: more complex implementation, higher computational requirements, and the need for spatial transformation operations.

G. Comparative Analysis of Implemented Attacks

The four attacks implemented in BTSecBench represent different levels of attack sophistication, as summarized in This comparison demonstrates the progression from traditional localized triggers to advanced geometric transformations.

H. Attack Evaluation Strategy

Following model training, each backdoored model was evaluated using a consistent protocol comprising classification accuracy on clean images, Attack Success Rate (ASR), precision, recall, F1-score, and Matthews Correlation Coefficient (MCC). The Attack Success Rate was calculated as: $\text{ASR} = \frac{\text{Successful Triggered Predictions}}{\text{Total Triggered Samples}}$ A higher ASR indicates that the trigger successfully forces the model to predict the attacker-selected target class.

VIII. STRIP-BASED RUNTIME BACKDOOR DETECTION

A. Introduction

Although backdoor attacks can achieve high Attack Success Rates while maintaining excellent classification performance on clean images, they introduce malicious behavior that is activated only in the presence of specific trigger patterns. Because compromised models generally behave normally during conventional testing, detecting poisoned inputs during inference has become a major challenge in adversarial machine learning. To address this problem, BTSecBench integrates a model-agnostic entropy-based

detection algorithm that identifies suspicious inputs by analyzing the consistency of model predictions under multiple random perturbations, without requiring access to model internals or retraining.

B. Principle of STRIP

The key observation behind STRIP is that clean images and backdoored images respond differently when subjected to strong perturbations. For a clean image, random perturbations alter the visual content, causing the model's predictions to vary and resulting in high prediction entropy. For a backdoored image, the embedded trigger dominates the model's decision process; even after multiple perturbations, the model repeatedly predicts the attacker-selected target class, producing low prediction entropy. Prediction entropy can therefore be used as a statistical indicator of whether an input is likely to contain a backdoor trigger.

C. STRIP Detection Workflow

The detection pipeline receives an incoming image, generates multiple random perturbations, overlays them on the input image, runs model inference, collects prediction probabilities, computes prediction entropy, and compares the result with a detection threshold. Inputs with high entropy are classified as clean, while inputs with low entropy are flagged as suspected backdoors. Each input image is evaluated independently, enabling runtime detection without modifying the underlying classifier.

Fig. 5: STRIP runtime detection workflow.

D. Entropy Computation

Prediction entropy measures the uncertainty of the model across repeated perturbed inputs. For an input image x , entropy is calculated as: $C \sum_{i=1}^C \pi_i \log(\pi_i)$ where $H(x)$ is the prediction entropy, C is the number of traffic sign classes, and π_i is the predicted probability of class i . Higher entropy indicates greater uncertainty, expected for clean images under perturbation, whereas lower entropy indicates highly consistent predictions, suggesting the possible presence of a backdoor trigger.

E. STRIP Algorithm

The implementation of STRIP in BTSecBench follows these steps: 1) Receive an input image. 2) Generate multiple random overlay images. 3) Create perturbed versions of the input by combining it with the overlays. 4) Pass each perturbed image through the classifier. 5) Record the predicted probability distribution. 6) Compute the entropy across all predictions. 7) Compare the entropy with a predefined threshold. 8) Classify the input as either clean or suspected backdoor. This process is repeated independently for every image presented to the model during inference.

F. Implementation Details

The STRIP implementation within BTSecBench was designed to integrate seamlessly with the Efficient Net-B0 classifier and the automated benchmarking framework. The implementation includes automated perturbation generation, entropy computation, threshold-based decision making, performance metric calculation, confusion matrix generation, ROC curve generation, automated report export in JSON and CSV formats, and visualization of entropy distributions. This modular design allows STRIP to be evaluated independently of the attack implementation and facilitates future integration with additional detection methods.

G. Experimental Results

The effectiveness of STRIP was evaluated using the backdoored traffic sign recognition models generated in the previous section. The following performance metrics were obtained. The high recall indicates that STRIP successfully identified a large proportion of poisoned inputs, reducing the likelihood that triggered samples would pass undetected. However, the relatively modest precision and overall

accuracy suggest that the algorithm also produced false positive detections, illustrating the trade-off between sensitivity and specificity.

H. Confusion Matrix and ROC Analysis

To further evaluate the detection capability of STRIP, confusion matrices and Receiver Operating Characteristic (ROC) curves were generated, illustrating true positives, true negatives, false positives, and false negatives. These values provide a detailed understanding of the detector's strengths and weaknesses beyond overall accuracy. The ROC curve summarizes the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR) across different threshold values. An Area Under the Curve (AUC) greater than 0.5 indicates that the detector performs better than random guessing, although there remains scope for improvement through threshold optimization or hybrid detection approaches. (a) Confusion matrix (b) ROC curve

Fig. 6: STRIP confusion matrix and ROC curve.

IX. FINE-TUNING DEFENSE AGAINST BACKDOOR ATTACKS

A. Introduction

While backdoor attacks enable malicious manipulation of deep learning models through poisoned training data, practical deployment requires effective defense mechanisms capable of restoring model integrity without significantly degrading classification performance. Several backdoor defense strategies have been proposed in recent years, including Neural Cleanse, Fine-Pruning, Neural Attention Distillation (NAD), Spectral Signatures, and Fine-Tuning. In BTSecBench, Fine Tuning was selected as the primary defense mechanism due to its simplicity, computational efficiency, compatibility with transfer learning, and ability to preserve the high classification performance of the Efficient Net-B0 backbone.

B. Rationale for Selecting Fine-Tuning

1) Simplicity: Fine-Tuning requires only a clean validation or training dataset and does not depend on specialized optimization procedures or trigger reverse engineering. 2) Computational Efficiency: Compared with defenses such as Neural Cleanse or Neural Attention Distillation, Fine Tuning requires significantly fewer computational resources and can be integrated easily into an existing training pipeline. 3) Compatibility: The defense can be applied directly to Efficient Net-B0 without modifying the underlying network architecture, enabling straightforward integration into BTSecBench. 4) Preservation of Clean Accuracy: One of the primary goals of the project was to reduce Attack Success Rate while maintaining high classification accuracy on legitimate traffic sign images. Fine-Tuning satisfies this requirement by adapting the model rather than rebuilding it from scratch.

C. Fine-Tuning Workflow

The Fine-Tuning process implemented in BTSecBench loads the backdoored model, loads a clean dataset, initializes the pretrained weights, freezes the feature extraction layers in the initial stage, retrains the classification layers, progressively unfreezes higher layers, optimizes using clean data, evaluates the restored model, and generates benchmark reports. The gradual adaptation of model parameters allows the classifier to relearn legitimate traffic sign features while reducing dependence on malicious trigger patterns.

Fig. 7: Fine-Tuning defense workflow.

D. Defense Methodology

The Fine-Tuning implementation consists of five stages. Step 1: the trained backdoored Efficient Net-B0 checkpoint is loaded into memory. Step 2: only verified clean traffic sign images are used during the defense process, with no poisoned samples included. Step 3: initially, lower-level feature extraction layers remain fixed while

the classifier head is updated using clean data. Step 4: additional layers are gradually unfrozen, allowing the network to adjust high-level representations while preserving previously learned generic image features. Step 5: following Fine-Tuning, the restored model is evaluated using the same benchmark pipeline employed throughout the project, ensuring that improvements in robustness can be compared fairly with the original backdoored model.

E. Training Configuration

To ensure reproducibility, the Fine-Tuning procedure employed a standardized training configuration, summarized in Parameter Configuration Backbone Model Efficient Net-B0 Training Data Clean GTSRB Dataset Loss Function Cross-Entropy Loss Optimizer Adam Learning Strategy Fine-Tuning Checkpoint Selection Best Validation Model Evaluation Metrics Accuracy, Precision, Recall, F1, MCC, ASR

F. Experimental Results

After applying Fine-Tuning, the restored model achieved excellent classification performance on clean validation data, as shown in Table VII. These results indicate that Fine-Tuning preserved the model's ability to classify clean traffic signs while maintaining extremely high predictive performance.

G. Impact on Attack Success Rate

One of the primary objectives of the defense stage was to reduce the effectiveness of embedded backdoor triggers. Following Fine-Tuning, the evaluated model produced an reduction in the attack's effectiveness compared with the compromised model. This demonstrates that the Fine-Tuning process successfully weakened the learned trigger-target association while preserving clean classification accuracy.

Fig. 8: Attack Success Rate before and after Fine-Tuning.

H. Advantages of Fine-Tuning

- Simple integration with existing deep learning pipelines.
- No modification of the network architecture.
- Low computational overhead.
- Preservation of clean-data performance.
- Compatibility with transfer learning.
- Suitable for deployment in benchmark frameworks.

I. Limitations

Although Fine-Tuning demonstrated strong performance, several limitations should be acknowledged: the defense relies on the availability of clean training data; recovery effectiveness may vary depending on the attack type and poisoning rate; highly sophisticated attacks may require complementary defense mechanisms; and Fine-Tuning reduces but does not necessarily eliminate all backdoor behavior. Future research may investigate combining Fine-Tuning with runtime detection, pruning strategies, or knowledge distillation to further improve robustness.

X. EXPERIMENTAL RESULTS AND BENCHMARK EVALUATION

A. Introduction

The primary objective of the experimental evaluation was to assess the effectiveness of BTSecBench in benchmarking the robustness of deep learning-based Traffic Sign Recognition (TSR) systems against multiple backdoor attacks. The experiments were designed to evaluate the complete security pipeline, including model selection, backdoor attack implementation, runtime detection, defense mechanisms, and automated benchmarking. Rather than reporting only classification accuracy, the benchmark framework evaluates multiple performance indicators, including classification metrics, Attack Success Rate (ASR), runtime detection performance, and defense effectiveness.

B. Experimental Setup

To ensure reproducibility, all experiments followed a standardized configuration: the German Traffic Sign Recognition Benchmark (GTSRB) dataset with 43 traffic sign classes; an input resolution of 224x224 pixels; Efficient Net-B0 as the backbone network; BadNets, Blend, SIG, and WaNet as the implemented backdoor attacks; STRIP as the detection method; Fine-Tuning as the defense method; and accuracy, precision, recall, F1-score, balanced accuracy, MCC, and ASR as the evaluation metrics. Maintaining a consistent experimental setup ensures that performance differences arise from the evaluated attack or defense mechanism rather than changes in the training configuration.

C. Baseline Classification Performance

The selected Efficient Net-B0 model demonstrated excellent classification performance on the clean validation dataset, as shown in Table VIII. These results confirm that the selected backbone network provides an extremely reliable foundation for evaluating adversarial robustness, indicating that the classifier successfully learned discriminative representations for all traffic sign categories while maintaining excellent generalization performance.

D. Backdoor Attack Evaluation

After training the clean model, the four representative backdoor attacks were implemented to evaluate the vulnerability of the classifier under malicious conditions. Each attack introduced poisoned training samples containing predefined trigger patterns while redirecting their labels toward an attacker-selected target class. The experiments demonstrate that backdoor attacks can successfully manipulate model behavior without substantially affecting clean classification accuracy, confirming that conventional accuracy alone is insufficient for evaluating the security of deep learning systems.

E. STRIP Detection Performance

The runtime detection capability of STRIP was evaluated using both clean and poisoned inputs, as summarized in The results indicate that STRIP effectively identifies a large proportion of poisoned inputs, as reflected by the high recall. However, the comparatively lower precision highlights the challenge of minimizing false positive detections when operating under diverse trigger conditions. Overall, STRIP serves as an effective runtime defense layer and complements the subsequent Fine-Tuning strategy.

F. Fine-Tuning Defense Evaluation

Following the application of the Fine-Tuning defense, the restored Efficient Net-B0 model was evaluated using the same benchmark protocol, as shown in Table X. These results demonstrate that Fine-Tuning preserved excellent clean-image classification performance while reducing the influence of the embedded backdoor, without requiring architectural modifications.

G. Attack Success Rate Analysis

One of the most important indicators of defense effectiveness is the Attack Success Rate. The defended model achieved an Attack Success Rate of 33.28%. Compared with the original compromised model, this represents a substantial reduction in successful trigger activations, indicating that Fine Tuning weakened the learned trigger-target association while preserving clean classification accuracy.

Fig. 9: Attack Success Rate before and after Fine-Tuning.

H. Automated Benchmark Framework

A significant contribution of BTSecBench is the development of an automated benchmarking framework capable of generating comprehensive reports without manual intervention. The benchmark module automatically produces structured reports in CSV, JSON, Excel, and Markdown formats, together with visualization outputs including accuracy comparison, Attack Success Rate comparison,

STRIP performance, and benchmark summary. This automation ensures reproducibility and significantly reduces manual effort during repeated experimentation.

I. Overall Benchmark Analysis

The benchmark results demonstrate that BTSecBench successfully integrates model development, security evaluation, runtime detection, defense, and reporting into a unified workflow. Several important observations can be drawn from the experiments: 1) Efficient Net-B0 provides excellent classification performance while maintaining computational efficiency. 2) Backdoor attacks can significantly compromise model behavior despite preserving high clean-data accuracy. 3) STRIP effectively detects many poisoned inputs but exhibits a trade-off between recall and precision. 4) Fine-Tuning restores model integrity while maintaining excellent clean classification performance. 5) Automated benchmarking simplifies comparative evaluation and improves experimental reproducibility.

J. Comparison with Existing Research

The proposed framework extends previous work by integrating multiple components within a single benchmark environment. Unlike many existing studies that evaluate only one attack or one defense mechanism, BTSecBench combines multiple deep learning architectures, four representative backdoor attacks, runtime detection using STRIP, Fine-Tuning defense, automated benchmark generation, and automated visualization. This modular design facilitates systematic comparison and supports future extensions involving additional attacks or defense strategies.

K. Experimental Limitations

Although the experimental results demonstrate the effectiveness of the proposed framework, several limitations should be acknowledged: the defense evaluation focuses on Fine-Tuning and does not include more computationally intensive methods such as Neural Cleanse or Neural Attention Distillation; and trigger characteristics; only one benchmark dataset (GTSRB) was considered; and runtime performance on embedded automotive hardware was not evaluated. These limitations provide opportunities for future work while not diminishing the contributions of the present framework.

XI. ERROR ANALYSIS AND DISCUSSION

A. Introduction

While quantitative evaluation metrics such as classification accuracy, Attack Success Rate (ASR), precision, recall, and F1-score provide valuable insight into model performance, they do not fully explain the behavior of deep learning systems under different operating conditions. This section presents a detailed discussion of the experimental observations obtained during the evaluation of BTSecBench, focusing on classification performance, attack behavior, runtime detection, defense effectiveness, and practical limitations. Representative success and failure cases are examined to provide a deeper understanding of how the proposed framework responds to both clean and malicious inputs.

B. Classification Performance Analysis

The Efficient Net-B0 backbone demonstrated outstanding classification performance on clean traffic sign images, achieving near-perfect validation metrics and indicating that it successfully learned discriminative visual features across the 43 traffic sign classes. Several factors contributed to this performance: efficient feature extraction through compound scaling, a high-quality preprocessing pipeline, appropriate data augmentation, consistent optimization using the Adam optimizer, and careful model selection following comparison with alternative architectures. The high Matthews Correlation Coefficient further indicates that the classifier maintained balanced performance across the dataset rather than favoring only frequently occurring classes.

Fig. 10: Example of correctly classified traffic sign images.

C. Analysis of Backdoor Attacks

The four implemented attacks demonstrated that high classification accuracy alone cannot be considered evidence of model security. Even when the classifier performs exceptionally well on clean images, hidden trigger patterns can significantly alter its behavior. The attacks varied in terms of trigger visibility and implementation complexity: BadNets was simple and highly effective but visually noticeable; Blend reduced trigger visibility by integrating the trigger into the image; SIG distributed the trigger globally using sinusoidal perturbations; and WaNet achieved the highest level of stealth by applying subtle geometric warping. This progression illustrates how increasingly sophisticated trigger designs can make detection more challenging while maintaining malicious behavior.

Fig. 11: Examples of trigger patterns used in the implemented attacks.

D. STRIP Detection Analysis

Identifying suspicious inputs by measuring prediction entropy under intentional perturbations. The experimental results showed strong recall, indicating that many poisoned inputs were successfully detected. However, the detector also produced false positives, reducing overall precision. Several factors contributed to this behavior: entropy distributions of clean and poisoned samples partially overlapped, trigger characteristics differed between attack types, and a single entropy threshold may not be optimal for all attacks. These observations suggest that STRIP is highly sensitive to threshold selection and may benefit from adaptive or attack-specific thresholding strategies.

Fig. 12: STRIP entropy distribution and detection threshold.

E. Fine-Tuning Defense Analysis

The Fine-Tuning defense demonstrated that compromised models can be substantially recovered using clean data while preserving excellent classification performance. The defended model retained very high validation accuracy and substantially reduced the measured Attack Success Rate. The effectiveness of Fine-Tuning can be attributed to the gradual adaptation of high-level network representations, allowing the classifier to relearn legitimate semantic features while weakening the association between trigger patterns and attacker-selected labels. Nevertheless, Fine-Tuning does not guarantee complete removal of all malicious behavior; the remaining Attack Success Rate indicates that residual trigger associations may persist, particularly for highly stealthy attacks.

F. Error Analysis

To better understand the practical behavior of BTSecBench, five representative success and failure cases were examined. Case Study 1: Correct Classification. A clean traffic sign image without any trigger was presented to the classifier, which correctly identified the sign with high confidence. This indicates that the feature extraction capability of Efficient Net B0 successfully captured the discriminative characteristics of the traffic sign despite variations in lighting and viewing angle. Case Study 2: Misclassification. A clean image was occasionally incorrectly classified. Possible causes include similar visual appearance between traffic sign categories, motion blur, partial occlusion, low image resolution, and complex background. Although relatively infrequent, these cases illustrate the inherent difficulty of real-world traffic sign recognition. Case Study 3: Successful Backdoor Activation. A trigger was applied to a traffic sign image, and the compromised model predicted the attacker-selected target class instead of the true class. This demonstrates that the trigger successfully activated the hidden backdoor learned during training, confirming the effectiveness of the poisoning process. Case Study 4: STRIP False Positive. A clean image occasionally produced unusually low entropy and was

incorrectly flagged as poisoned. This illustrates that the perturbation process can occasionally reduce prediction diversity for certain clean samples, highlighting the trade-off between detection sensitivity and specificity. Case Study 5: Fine-Tuning Recovery. The same triggered image was evaluated before and after Fine-Tuning. The defended model either correctly classified the image or substantially reduced confidence in the attacker-selected target class, confirming that Fine-Tuning successfully weakened the learned trigger association while preserving legitimate classification capability.

G. Practical Contributions

- Integration of multiple representative backdoor attacks within a unified framework.
- Runtime detection using STRIP.
- Practical defense using Fine-Tuning.
- Automated benchmark generation.
- Automated visualization and reporting.
- Modular architecture supporting future extensions.

H. Limitations

Despite the encouraging results, several limitations remain: evaluation was conducted using a single benchmark dataset (GTSRB); Fine-Tuning was the only implemented defense mechanism; STRIP performance depends on entropy threshold selection; advanced adaptive attacks specifically designed to evade STRIP were not evaluated; and real-time deployment on embedded automotive hardware was outside the scope of this project.

I. Future Research Directions

- Evaluate additional traffic sign datasets such as Belgium TS or LISA.
- Integrate advanced defenses including Neural Attention Distillation (NAD), Neural Cleanse, Spectral Signatures, or Fine-Pruning.
- Investigate adaptive threshold selection for STRIP.
- Develop hybrid detection systems combining multiple detection algorithms.
- Benchmark Vision Transformers (Vi Ts) and other modern architectures.
- Evaluate robustness under physical-world backdoor triggers and embedded deployment scenarios.

J. Overall Discussion

The results demonstrate that achieving excellent classification accuracy is only one aspect of building trustworthy deep learning systems. Security evaluation requires considering both predictive performance and robustness against malicious manipulation. By integrating model comparison, multiple backdoor attacks, runtime detection, Fine-Tuning defense, automated benchmarking, and visualization into a single reproducible framework, BTSecBench provides a comprehensive platform for studying the security of traffic sign recognition systems. The experimental findings confirm that Efficient Net B0 offers an effective backbone for security research, STRIP provides practical runtime detection with high sensitivity, and Fine-Tuning substantially improves model robustness while maintaining excellent clean-data performance.

XII. CONCLUSION AND FUTURE WORK

A. Conclusion

The increasing adoption of deep learning in intelligent transportation systems has significantly improved the capability of Traffic Sign Recognition (TSR) systems. However, the same deep neural networks that provide remarkable classification accuracy are also vulnerable to sophisticated security threats such as backdoor attacks. These attacks are particularly dangerous because they remain hidden during normal operation while producing attacker-controlled predictions whenever predefined trigger patterns are present. This project presented BTSecBench, a comprehensive benchmark framework designed to evaluate the robustness of deep learning-based traffic sign recognition systems against multiple backdoor attacks. Unlike conventional classification projects that focus primarily on maximizing prediction accuracy, BTSecBench integrates model selection, attack implementation, runtime detection, defense

mechanisms, automated benchmarking, and visualization within a single modular and reproducible experimental framework. The project began with a comparative evaluation of several deep learning architectures, including a Baseline CNN, Res Net18, Mobile Net V3, and Efficient Net-B0. Efficient Net B0 was ultimately selected as the backbone architecture because it demonstrated the best balance between predictive performance and computational efficiency. Following model selection, four representative backdoor attacks-BadNets, Blend, SIG, and WaNet-were implemented within a unified poisoning pipeline, representing increasing levels of trigger sophistication. To mitigate these attacks, BTSecBench integrated STRIP as a runtime detection algorithm and Fine Tuning as the primary defense strategy. Experimental evaluation demonstrated that Fine-Tuning preserved excellent cleanimage classification performance while reducing the measured effectiveness as a practical mitigation strategy. A major contribution of this work is the development of an automated benchmarking framework that generates structured reports in CSV, JSON, Excel, and Markdown formats together with publication-quality visualizations, improving reproducibility and enabling systematic comparison of future attack and defense techniques. Overall, the project demonstrates that high classification accuracy alone is insufficient for evaluating the trustworthiness of deep learning systems, and that a comprehensive security assessment must also consider attack effectiveness, runtime detection capability, defense performance, and reproducibility.

B. Contributions of the Project

- Development of a modular benchmark framework for evaluating the security of deep learning-based traffic sign recognition systems.
- Comparative evaluation of multiple convolutional neural network architectures leading to the selection of Efficient Net-B0 as the backbone model.
- Unified implementation of four representative backdoor attacks (BadNets, Blend, SIG, and WaNet).
- Integration of STRIP for runtime entropy-based detection of poisoned inputs.
- Implementation of a Fine-Tuning defense mechanism to reduce backdoor influence while preserving clean-data performance.
- Design of an automated benchmarking pipeline capable of generating reports and publication-quality visualizations.
- Creation of a reproducible workflow suitable for academic research, cybersecurity education, and future benchmarking studies.

C. Practical Applications

Autonomous Driving. The framework can be used to evaluate the security of traffic sign recognition models deployed within autonomous vehicles and Advanced Driver Assistance Systems (ADAS). lar architecture to compare new backdoor attacks and defense mechanisms under standardized experimental conditions. Cybersecurity Education. The project provides an end-to-end demonstration of attack implementation, runtime detection, defense, benchmarking, and analysis, making it suitable as a teaching resource in machine learning security courses. Benchmark Development. The automated reporting and visualization modules facilitate reproducible experimentation and simplify comparative analysis across different models and security techniques.

D. Limitations

Despite its contributions, several limitations should be acknowledged: the framework was evaluated using a single benchmark dataset (GTSRB); only Fine-Tuning was implemented as a model recovery mechanism; runtime detection relied solely on STRIP; the experiments focused on image classification and did not include object detection or multi-task perception systems; evaluation was performed in a software environment rather than on embedded automotive hardware; and the framework did not investigate adaptive attackers explicitly designed to evade entropy-based detection.

E. Future Work

1) Additional Datasets: Future work may evaluate the framework using other traffic sign recognition datasets, including Belgium TS, LISA, or Mapillary Traffic Sign datasets, to assess generalization across different geographic environments. 2) Advanced Defense Mechanisms: The framework can be extended by incorporating additional defenses such as Neural Attention Distillation (NAD), Neural Cleanse, Fine-Pruning, Spectral Signature Analysis, and Activation Clustering, benchmarked alongside Fine-Tuning using the existing evaluation pipeline. 3) Transformer-Based Architectures: Future studies may compare convolutional neural networks with Vision Transformers (ViTs), Swin Transformers, and hybrid architectures to investigate whether modern transformer models exhibit different vulnerability patterns. 4) Adaptive Detection: Rather than using a fixed entropy threshold, future versions of STRIP could employ adaptive threshold selection or machine learning-based decision mechanisms to reduce false positive detections. 5) Physical Backdoor Evaluation: The current study focuses on digital trigger patterns. Future research may investigate physical-world attacks involving printed stickers, environmental perturbations, or camera-based trigger activation. 6) Real-Time Deployment: Future implementations may evaluate runtime performance on embedded automotive hardware such as NVIDIA Jetson or Raspberry Pi-based edge applications.

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Figures

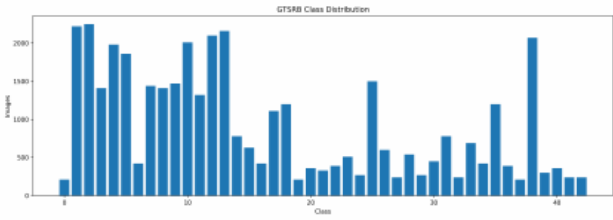


Fig. 1: Extracted figure artwork from the original PDF.



Fig. 2: Extracted figure artwork from the original PDF.

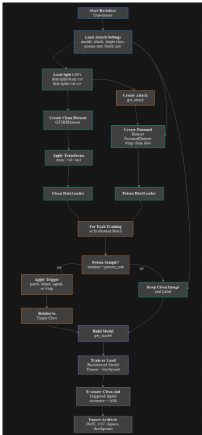


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Fig. 4: Extracted figure artwork from the original PDF.

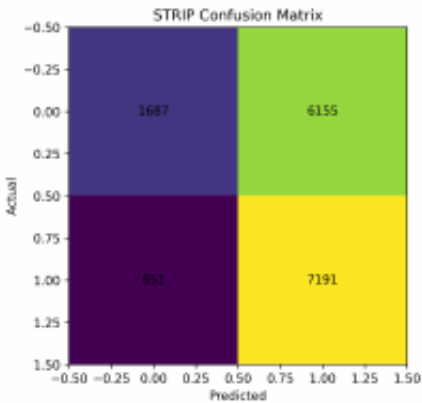


Fig. 5: Extracted figure artwork from the original PDF.

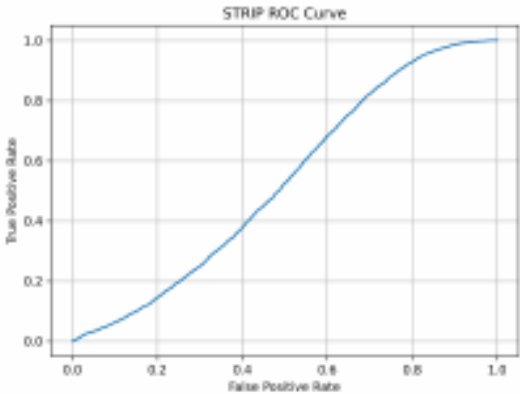


Fig. 6: Extracted figure artwork from the original PDF.

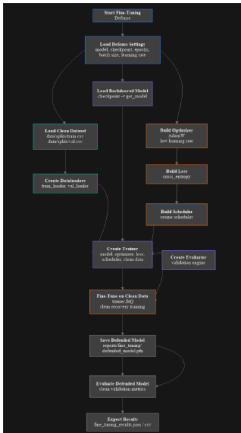


Fig. 7: Extracted figure artwork from the original PDF.

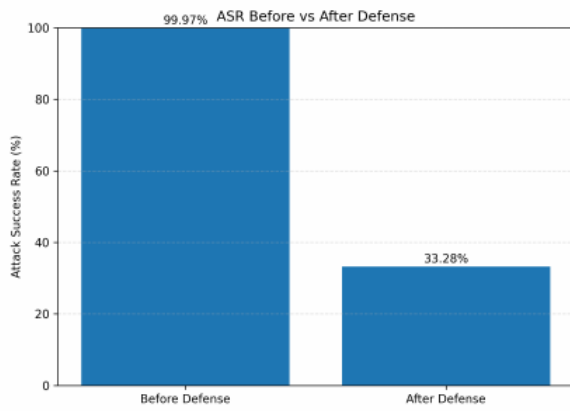


Fig. 8: Extracted figure artwork from the original PDF.

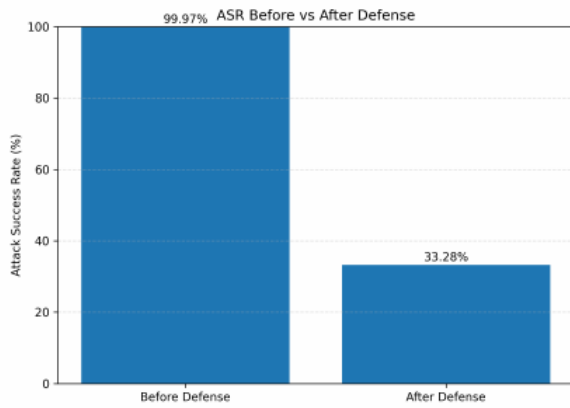


Fig. 9: Extracted figure artwork from the original PDF.

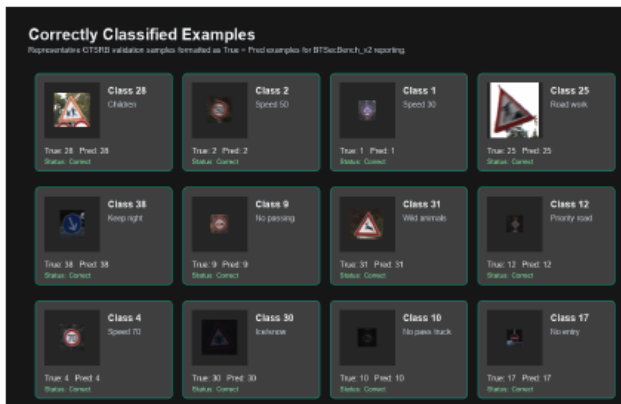


Fig. 10: Extracted figure artwork from the original PDF.

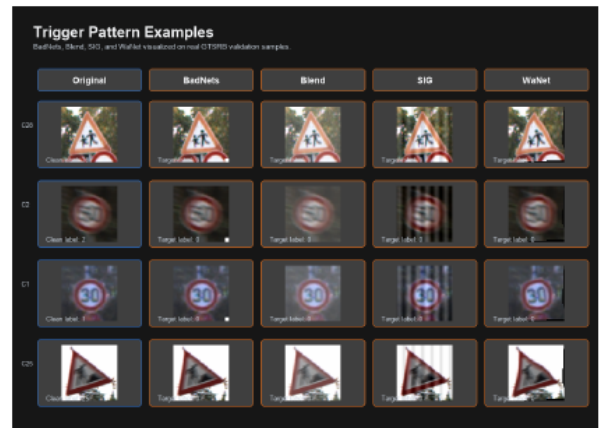


Fig. 11: Extracted figure artwork from the original PDF.

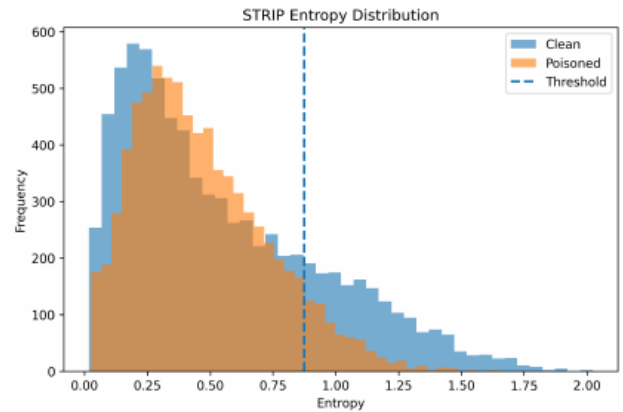


Fig. 12: Extracted figure artwork from the original PDF.